

Electrochemical equilibrium from reaction thermodynamics

Read and prepare: 15–20 minutes total. Bring tentative answers; uncertainty is useful for discussion.

What determines the sign of a reversible concentration-cell potential?

Reaction direction and electrons

Write both half-reactions as reductions. Reverse the left half-reaction and combine it with the right after balancing electrons. Define $E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}$. Both standard reduction potentials need the same reference electrode and temperature. Scaling a half-reaction changes amounts and electron count, not its potential.

Activities connect chemistry to voltage

Electrochemical potential includes chemical and electrical contributions. For the balanced cell reaction, $\Delta_r G = -nFE$ and $\Delta_r G^\circ = -nFE^\circ$. Combining these with $\Delta_r G = \Delta_r G^\circ + RT \ln Q$ gives the Nernst equation. Activities are dimensionless and belong to their compartments. Concentration approximates activity only under stated assumptions.

A reversible prediction

The calculated potential concerns equilibrium driving force under the chosen conventions. Junction potentials, kinetic losses and resistive drops can affect an actual measured or loaded cell. Keep temperature fixed unless the standard-potential data remain applicable. A reaction quotient from an arbitrary state is not automatically an equilibrium constant.

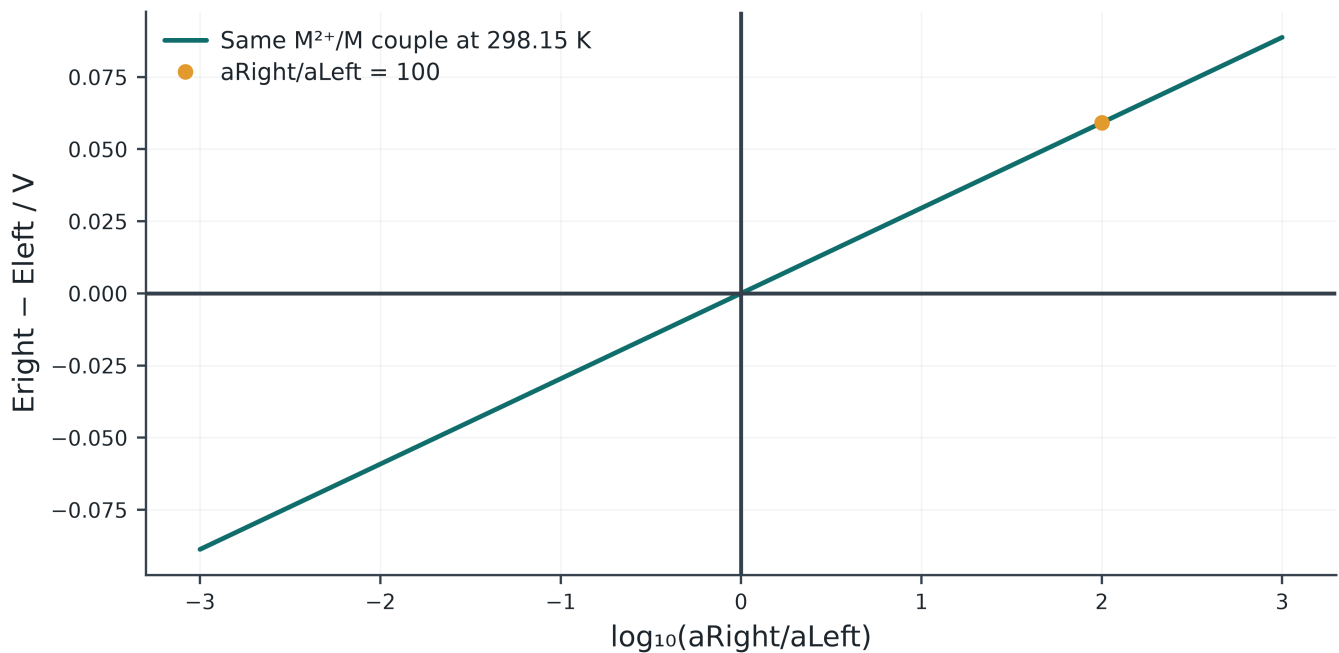
$$E = E^\circ - \frac{RT}{nF} \ln Q; \Delta_r G = -nFE$$

$$\ln K = \frac{nFE^\circ}{RT}; R = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}$$

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

For the same M^{2+}/M reduction couple on both sides at 298.15 K, take $a_{\text{Left}}=0.0100$ and $a_{\text{Right}}=1.000$, $n=2$ and equal standard potentials. The reaction as written transfers the dissolved species from the right compartment to the left, so $Q=a_{\text{Left}}/a_{\text{Right}}=0.0100$. With $E^\circ=0$, $E_{\text{right}}-E_{\text{left}}=+0.05915935$ V. Using $F=96485.33212$ C/mol gives $\Delta rG \approx -11.416$ kJ per mole of the written cell reaction.



Check and interpret

Exchange the two activities: E reverses sign. Set them equal: $E=0$. Double the complete reaction basis: n and ΔrG double, while E remains unchanged. This scaling check tests whether the reaction basis and electron count have been used consistently.

Bring three short answers

A sketch, a few numbers or one sentence is enough. Do this before opening the calculator; revise your prediction in class.

1. Predict the sign of E before calculating when the right activity exceeds the left.

2. Why can E be nonzero even though $E^\circ=0$ for this concentration cell?

3. Which quantities change when the complete reaction and electron count are doubled?

Optional extension

Use the activity-model controls only within their stated domains, or relate a fuel-cell equilibrium voltage to reaction Gibbs energy. Separate reversible potential from operating voltage. The supplied table of contents locates fuel-cell voltage in 17.6; the present handout develops the concentration-cell bridge independently.

Continue in the core lab

<https://www.skhgroup.net/teaching/2105603/labs/electrochemistry.html>

Reading locator: supplied textbook contents, sections 11.11, 14.8-14.9, 17.6.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. Required weekly independent work: 15–20 minutes preparation plus 30–40 minutes follow-up; optional extensions are additional.